Issue date: 12/1/2003
Revision date: 10/20/2021

Safety Data Sheet

1. Chemical product and company identification

Product name : Biphenyl

Company information

Name of manufacturer : KANTO CHEMICAL CO., INC.
Address : 2-1, Nihonbashi, Muromachi 2-Chome, Chuo-Ku, Tokyo, 103-0022, JP
Name of section : Business Administration Department, Reagent Division
Telephone number : +81-3-6214-1090
Facsimile number : +81-3-3241-1047
Mail address : BC32@kanto.co.jp
Reference No : 10429

2. Hazards identification

GHS classification

Health hazards	Serious eye damage/eye irritation	Category 2B
	Carcinogenicity	Category 1B
	Specific target organ toxicity (repeated exposure)	Category 1 (liver、nerve system、respiratory organs)
	Specific target organ toxicity (repeated exposure)	Category 2 (kidney)
Environmental hazards	Aquatic acute	Category 1

Hazard pictograms





Signal word : Danger

Hazard statements : Causes eye irritation
May cause cancer
Causes damage to organs (liver、nerve system、respiratory organs) through prolonged or repeated exposure
May cause damage to organs (kidney) through prolonged or repeated exposure
Very toxic to aquatic life

Precautionary statements

Prevention : Do not handle until all safety precautions have been read and understood.
Do not breathe dust.
Wash hands, forearms and face thoroughly after handling.
Do not eat, drink or smoke when using this product.
Avoid release to the environment.
Wear protective gloves/protective clothing/eye protection/face protection.

Response : IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue

		rinsing. IF exposed or concerned: Get medical advice/attention. Get medical advice/attention if you feel unwell. If eye irritation persists: Get medical advice/attention. Collect spillage.
Storage	:	Store locked up.
Disposal	:	Dispose of contents/container to hazardous or special waste collection point, in accordance with local, regional, national and/or international regulation.

3. Composition/information on ingredients

Distinction of substance or mixture : Substance

Synonyms : Diphenyl

Chemical name	Concentration (%)	Formula	TSCA	EC-No.	CAS RN
Biphenyl	≥ 99	C12H10	Listed	202-163-5	92-52-4

4. First aid measures

First aid measures

First-aid measures after inhalation : Remove the victim to fresh air, and make him blow his nose and gargle.

First-aid measures after skin contact : Wash the affected areas under running water.

First-aid measures after eye contact : Wash the affected areas under running water for at least 15 minutes. If necessary, get medical treatment.

First-aid measures after ingestion : Give the victim water or salt water and make him vomit. Get medical attention.

Personal Protection in First Aid and Measures : Rescuers should wear proper protective equipment like rubber gloves, goggles.

Most Important Symptoms/Effects

Symptoms/effects : If inhaled, causes cough, nausea, and vomiting.

5. Fire fighting measures

Suitable extinguishing media : Dry chemical powder, carbon dioxide, dry sand, foam

Unsuitable extinguishing media : Water spray

Fire hazard : If state of powdered, may cause dust explosion.

Firefighting instructions : Move containers from fire area if it can be done without risk, if not possible, apply water from a safe distance to cool and protect surrounding area.

Fight fire from windward.

Dry chemical powder, carbon dioxide or dry sand should be used for small fires. Foam extinguisher is effective for a large scale fire.

Personal protection (Emergency response) : Wear breathing apparatus.

6. Accidental release measures

Personal Precautions, Protective Equipment and Emergency Procedures

General measures	:	Wear proper protective equipment and avoid contact with skin and inhalation of dust. Conduct operations from upwind and evacuate people downwind.

Environmental precautions

Environmental precautions	:	Attention should be given to avoid damage to the environment by flowing of spillage to rivers.

Methods and Equipment for Containment and Cleaning up

For containment	:	Sweep up in a chemical waste container. Flush contaminated area with copious amounts of water.
Prevention Measures for Secondary Accidents	:	Remove nearby sources of ignition and prepare extinguishing media.

7. Handling and storage

Handling

Technical measures	:	Wear appropriate protective equipment to avoid contact with skin or inhalation of dust.
Precautions for safe handling	:	Avoid formation of dust and aerosols.

Storage

Storage conditions	:	Store in a dark, cool place and tightly closed.
Material used in packaging/containers	:	Glass, polyethylene, polypropylene.

8. Exposure controls / Personal protection equipment

ACGIH TWA	0.2 ppm

Appropriate engineering controls	:	Install a local ventilation system in case of dusty condition.

Protective equipment

Respiratory protection	:	If necessary, wear dust mask
Hand protection	:	Impervious protective gloves
Eye protection	:	Safety goggles
Skin and body protection	:	Protective clothing, protective boots

9. Physical and chemical properties

Physical state	:	Solid
Color	:	White
Odor	:	Aromatic
pH	:	No data available
Melting point	:	69 - 72 ° C
Freezing point	:	No data available
Boiling point	:	255.2 ° C
Flash point	:	113 ° C (C.C.)
Auto-ignition temperature	:	540 ° C

Decomposition temperature	:	No data available
Flammability (solid, gas)	:	Flammable
Vapor pressure	:	1.19 Pa (25℃)
Relative density	:	No data available
Density	:	No data available
Relative gas density	:	5.31
Solubility	:	Water; Insoluble. Organic solvents; Soluble in diethyl ether, dioxane.
Partition coefficient n-octanol/water (log Pow)	:	3.98
Explosive limits (vol %)	:	0.6 - 5.8 vol %
Viscosity, kinematic:	:	No data available
Particle characteristics	:	No data available

10. Stability and reactivity

Reactivity	:	May react with oxidizing substances.
Chemical stability	:	Stable under normal conditions.
Possibility of hazardous reactions	:	Stable under normal conditions of use.
Conditions to avoid	:	Light, heat.
Incompatible materials	:	Oxidizing substances.
Hazardous decomposition products	:	Carbon monoxide.

11. Toxicological information

Acute toxicity (oral)	:	No classification rat LD50=2400mg/kg
Acute toxicity (dermal)	:	No classification rabbit LD50=2500mg/kg
Acute toxicity (inhalation)	:	No classification (gas) Classification not possible (vapor) Classification not possible (dust, mist)
Skin corrosion/irritation	:	Classification not possible Both rabbit tests and dermal application tests with humans found the substance not irritating to skin. These results suggested that this substance falls under "No classification" category. However, it is reported to irritate skin according to the Initial Risk Assessment Report, and its EU Risk Phrase is R36/37/38, which indicates that the substance is a mild skin irritant. Thus, the substance was classified into the "Classification not possible" category.
Serious eye damage/irritation	:	Causes eye irritation Since the substance was slightly irritating when applied to rabbits' eyes in an eye irritation test, it was classified into category 2B.
Respiratory sensitization	:	Classification not possible
Skin sensitization	:	No classification In a guinea pig maximization test, there was reportedly no evidence of a skin sensitizing potential for this substance. Thus, the substance was classified into the "Not classification" category.

Germ cell mutagenicity	:	No classification Since an in vivo mutagenicity test using somatic cells (chromosomal aberration test using rat bone marrow cells) gave negative results, the substance was classified into the "No classification" category.
Carcinogenicity	:	May cause cancer In a carcinogenicity test in rats (2-year diet administration), an increased incidence of urinary bladder tumors was observed in a male dosed group. In dosed groups, calculus in the pelvis and urinary bladder and hyperplasia in the urinary bladder were observed in both males and females, but an increased incidence of tumors was not observed in the urinary bladder of females. Morever, in a carcinogenicity test in mice (2-year diet administration), an increased incidence of hepatocellular carcinoma was observed in female dosed groups. Frome the above, it was classified into category 1B.
Reproductive toxicity	:	No classification In a rat reproduction test, no effects of the substance on fertility were detected. In addition, in an oral administration test using pregnant rats, no serious reproductive toxicity was detected at doses that cause general toxic effects on the parental animals. Based on these reports, the substance was classified into the "No classification" category.
STOT-single exposure	:	Classification not possible In inhalation exposure tests using mice, mild respiratory discomfort were noted. Nevertheless, due to lack of sufficient test results, the substance was classified into the "Classification not possible" category.
STOT-repeated exposure	:	Causes damage to organs (liver、nerve system、respiratory organs) through prolonged or repeated exposure May cause damage to organs (kidney) through prolonged or repeated exposure In occupational exposure cases related to this substance, liver damage, effects on the central and peripheral nervous systems, and bronchitis were reported. Thus, the substance was classified into category 1 (liver, nervous system, respiratory organs). In addition, in a 21-day mixed diet administration test using rats, effects on the kidney was noted at the doses that fall under the range of category 2 guidance values. Thus, the substance was classified into category 2 (kidney).
Aspiration hazard	:	Classification not possible

12. Ecological information

Ecotoxicity

Aquatic acute	:	Very toxic to aquatic life Daphnia magna LC50=360μg/L/48h
Aquatic chronic	:	No classification

Persistence and degradability

Readily biodegradable
BOD : 66%

Bioaccumulative potential

Low bioconcentration
BCF : 280

Mobility in soil

Low mobility
Koc : 870-3300

Hazardous to the ozone layer

Ozone : Classification not possible

13. Disposal considerations

Ecology - waste materials : Mixed with flammable organic solvents and burn in a chemical incinerator equipped with an afterburner and a scrubber. Or entrust approved waste disposal companies with the disposal.

Contaminated container and packaging : In case of disposal of empty bottles, dispose bottles after removing the content thoroughly.

14. Transport information

International Regulations

Transport by sea(IMDG)

UN-No. (IMDG) : 3077
Proper Shipping Name (IMDG) : ENVIRONMENTALLY HAZARDOUS SUBSTANCE, SOLID, N.O.S.
Packing group (IMDG) : III
Transport hazard class(es) (IMDG) : 9

Air transport(IATA)

UN-No. (IATA) : 3077
Proper Shipping Name (IATA) : Environmentally hazardous substance, solid, n.o.s.
Packing group (IATA) : III
Transport hazard class(es) (IATA) : 9

Marine pollutant : Applicable

Transport in bulk according to Annex II of MARPOL 73/78 and the IBC Code

Pollutant category : X
MFAG-No : 171

15. Regulatory information

Regulatory information with regard to this substance in your country or region should be examined by your own responsibility.

16. Other information

Data sources : Solvents Handbook, T, Asahara el, Kodansha Scientific Ltd.(1976).
Handbook of Dangerous Substances Springer-Verlag Tokyo(1991).
Handbook of 17221 Chemical Products, The Chemical Daily Co.(2021).
NITE Chemical Risk Information Platform (NITE-CHRIP), National Institute of Technology and Evaluation.

The information contained herein is based on several references and the present state of our knowledge. However the SDS does not always cover all information about the product, handle the product carefully.The information is intended to ordinary usage, in case of particular handlings, conduct appropriate safety measurements.The information herein is only provision of information, and it does not represent a guarantee the properties of the product.The Safety Data Sheet(SDS) is prepared based on JIS Z7253.